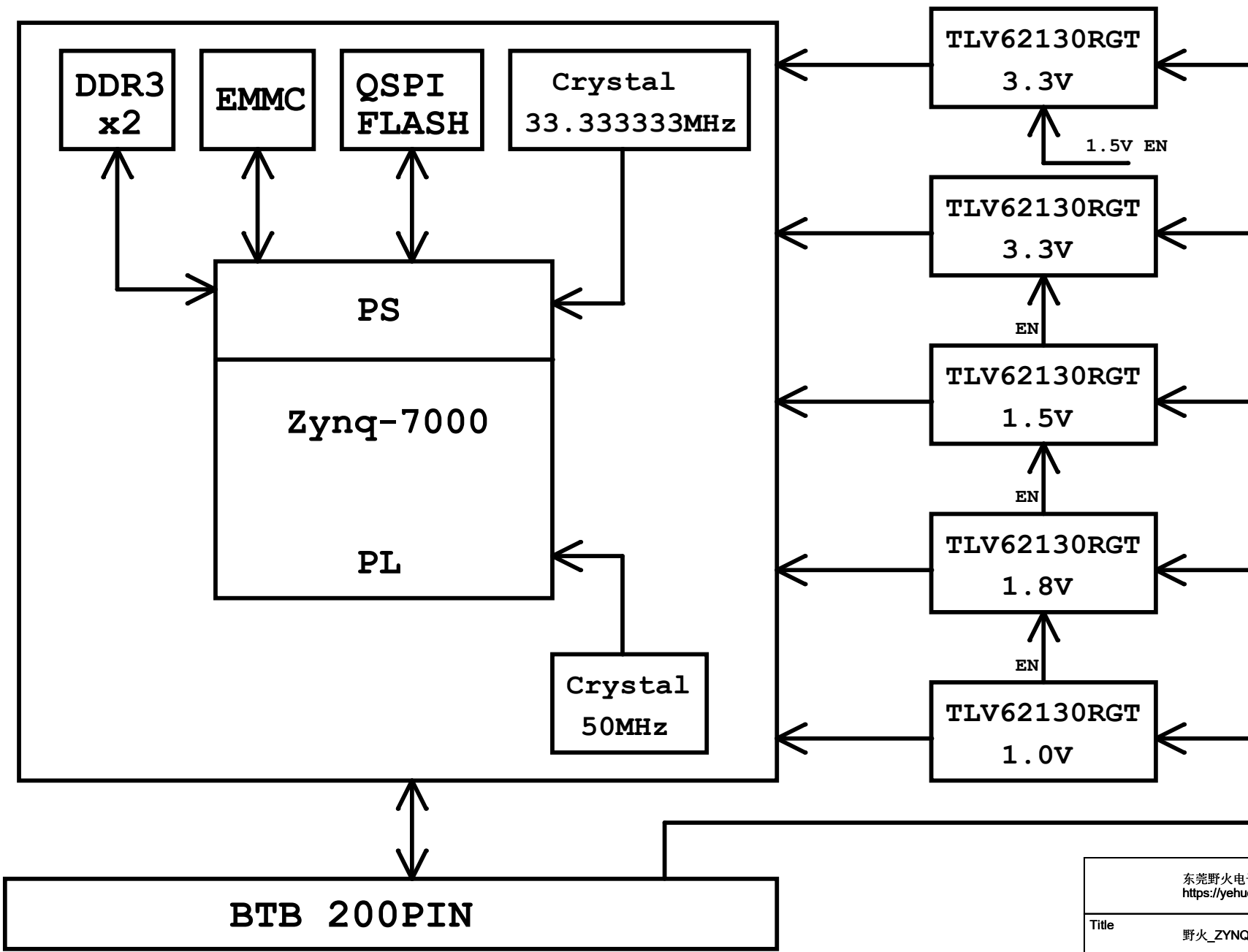


目录

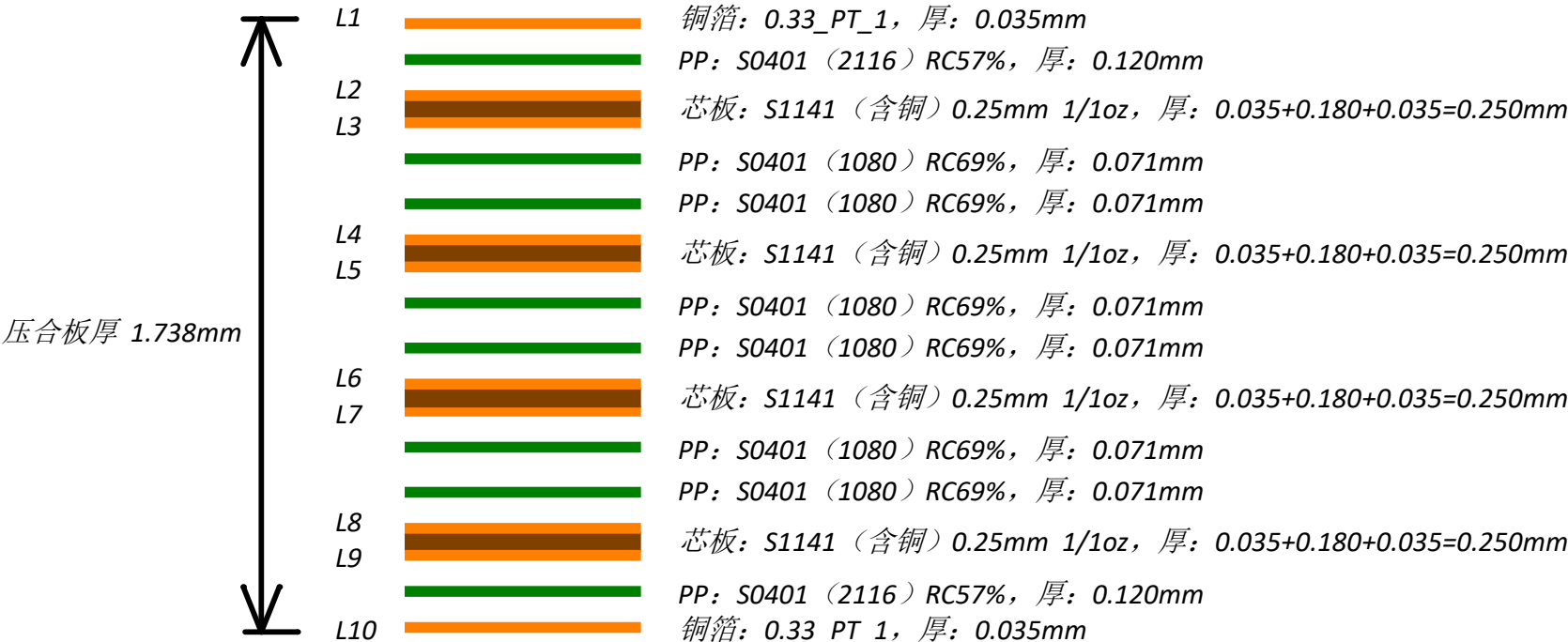
Page 1	目录
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Page 5	Power IN
Page 6	BTB Connector
Page 7	Bank0
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Page 14	BOOT

版本号	日期	设计	描述
V1.0	2020-10-26	xgh	初始版本
V1.1	2023-06-26	GHS	BTB初始版本
V1R2	2025-03-07	GHS	修正电源供电芯片的PG开启信号错误问题。



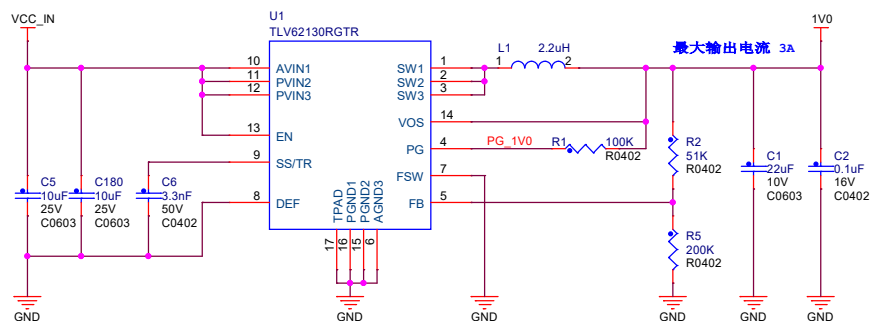
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ebf_xc7z010/020 core board

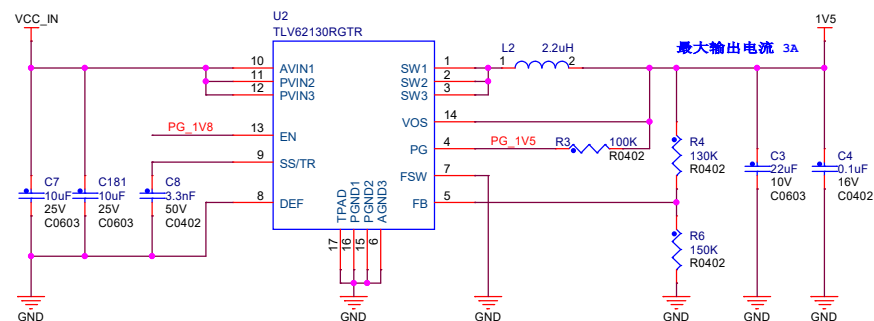


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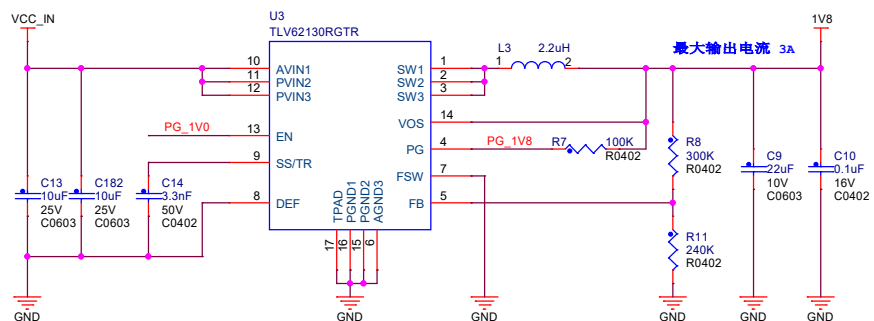
POWER 1.0V



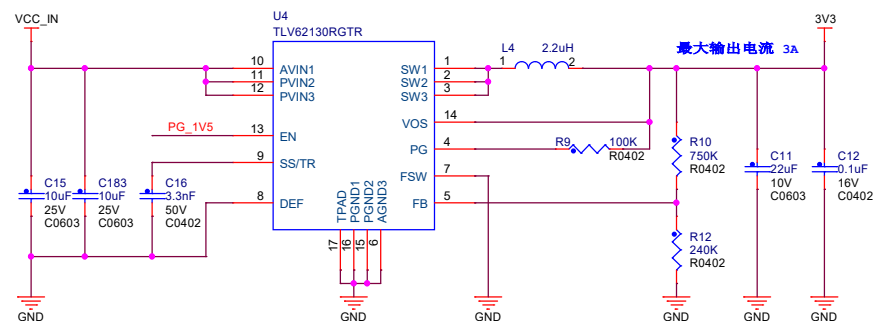
POWER 1.5V



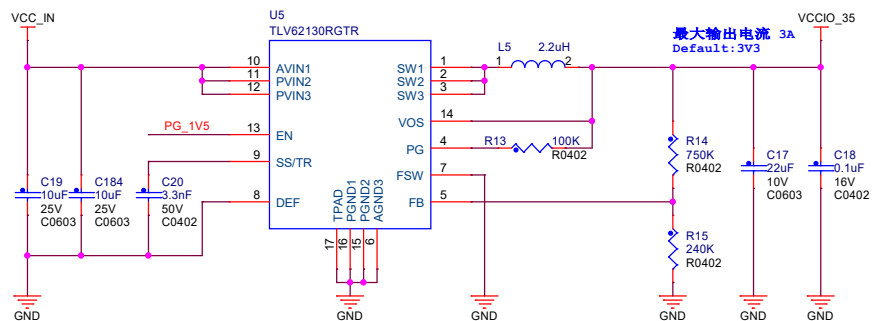
POWER 1.8V



POWER 3.3V



POWER VCCIO_35



Power IN Voltage 5V~12V

Open sequence

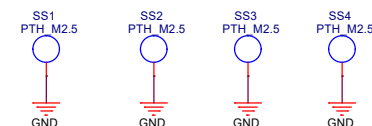
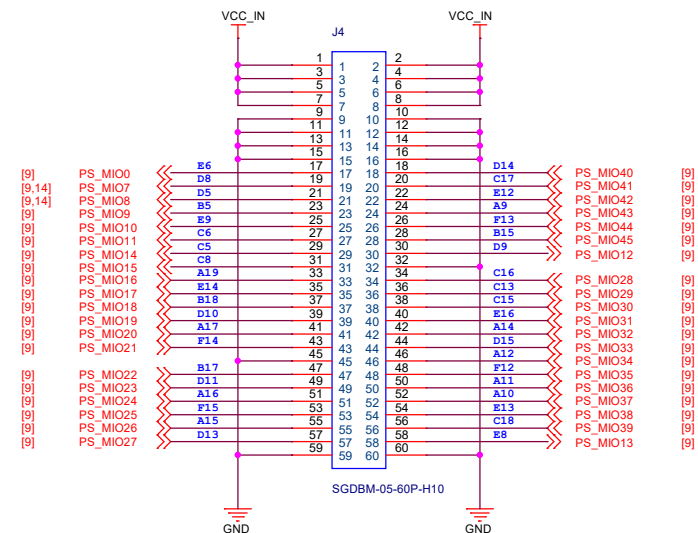
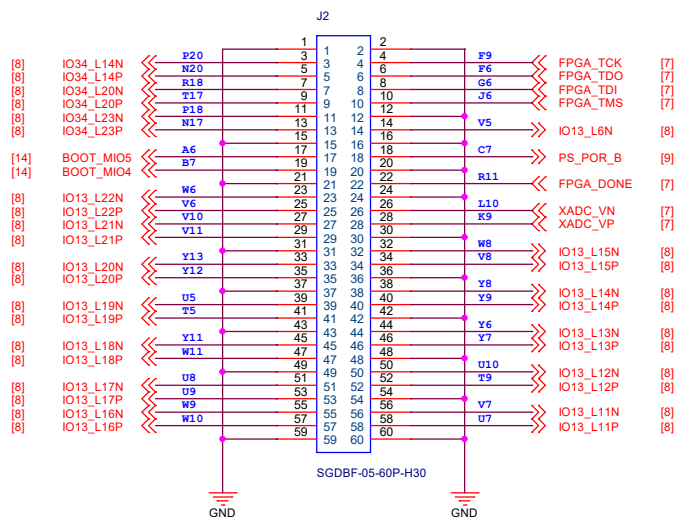
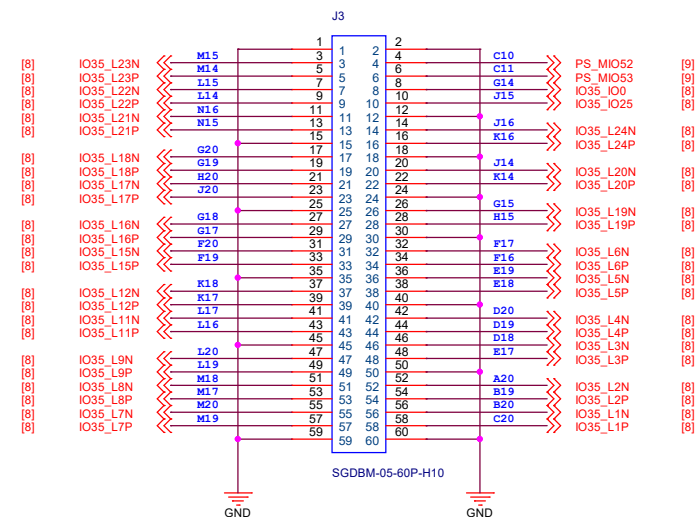
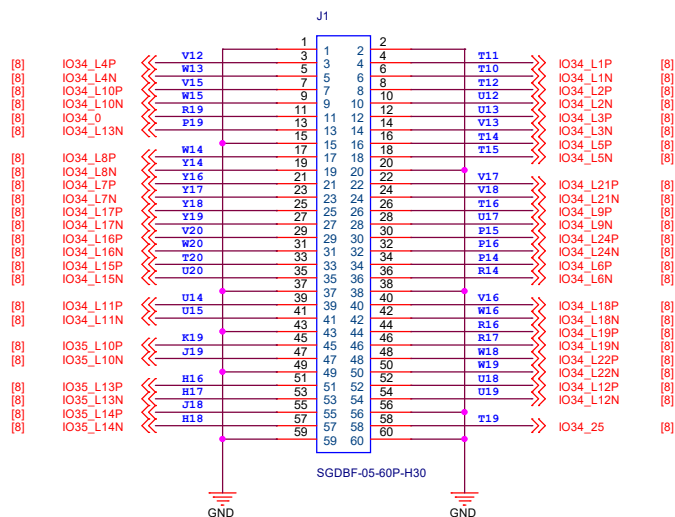
1.0V->1.8V->1.5V->3.3V->VCCIO_35

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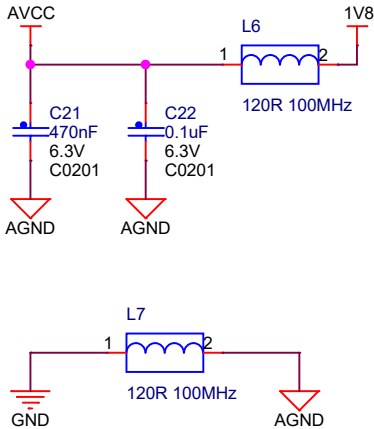
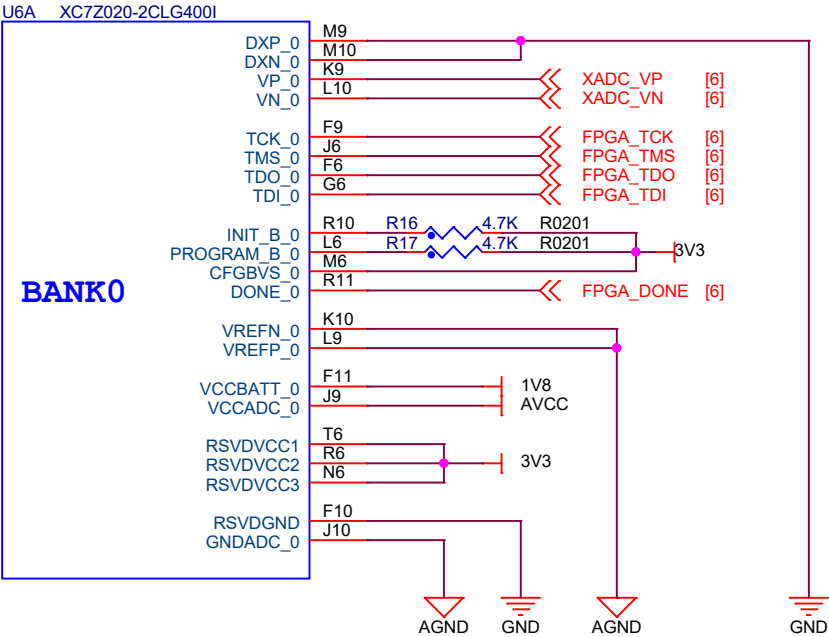
BTB Connector

60*4个引脚包含

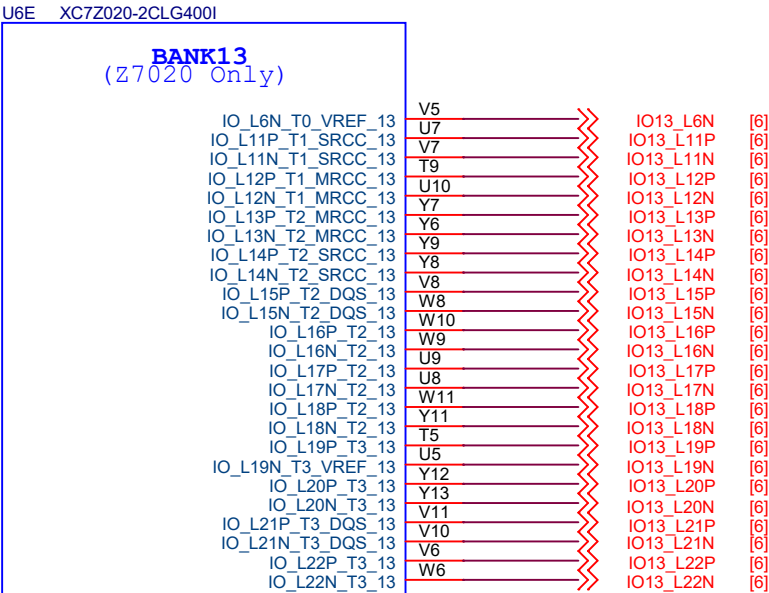
XADC	2
POR	1
DONE	1
PS_IO	42
PL_IO	12
JTAG	4
12V输入	8
GND	55
BOOT	2



BANK0

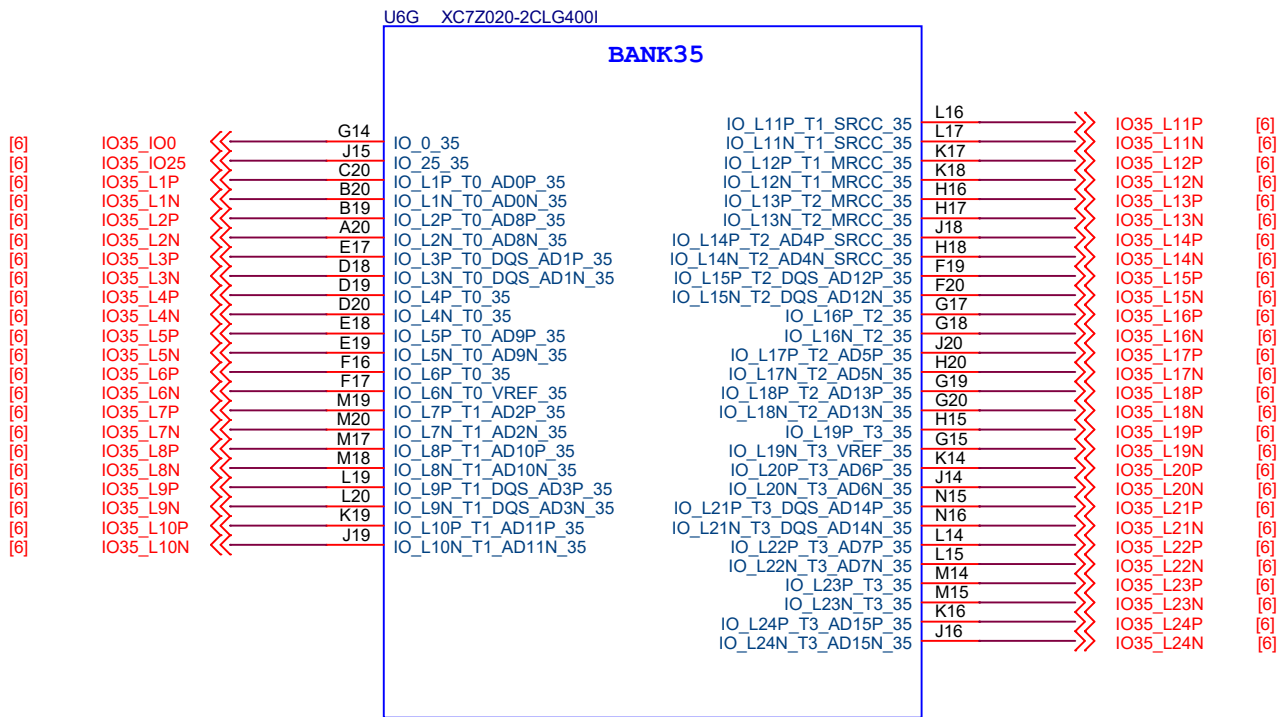
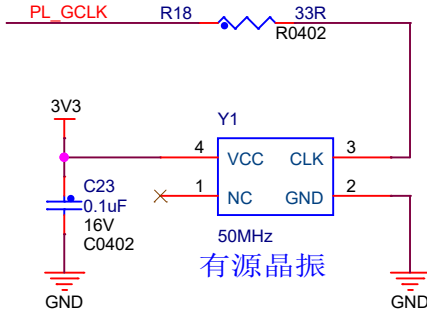
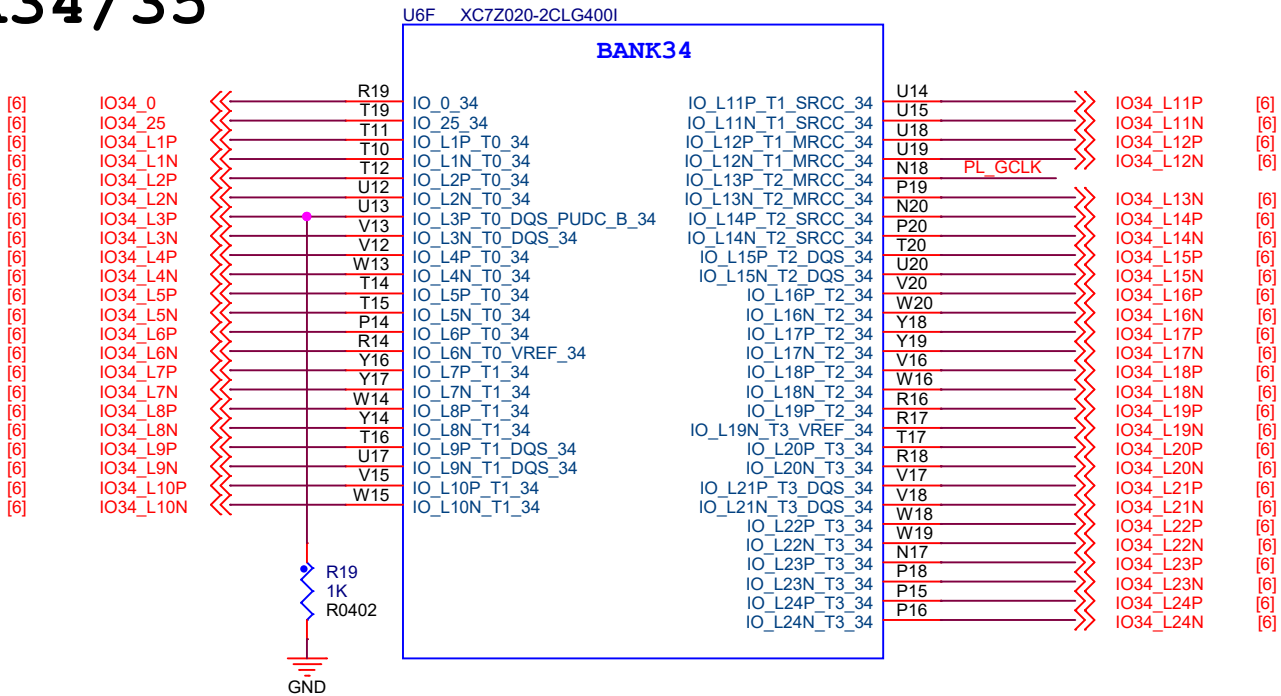


BANK13



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BANK34/35

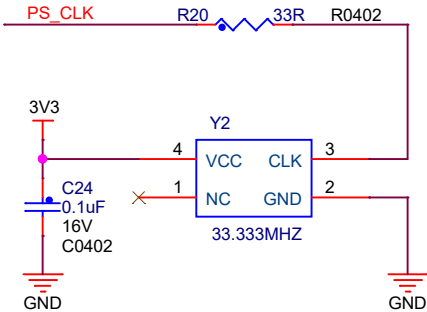
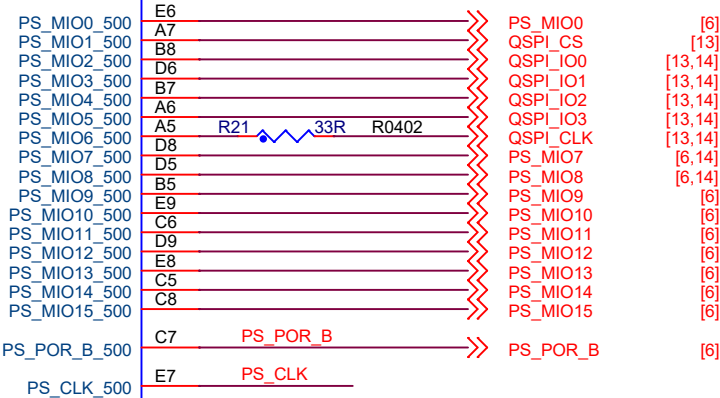


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BANK500/501

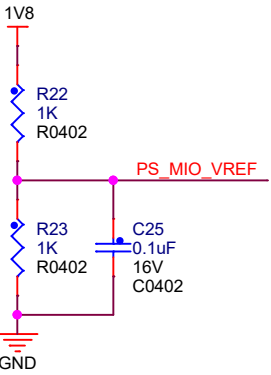
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BANK500



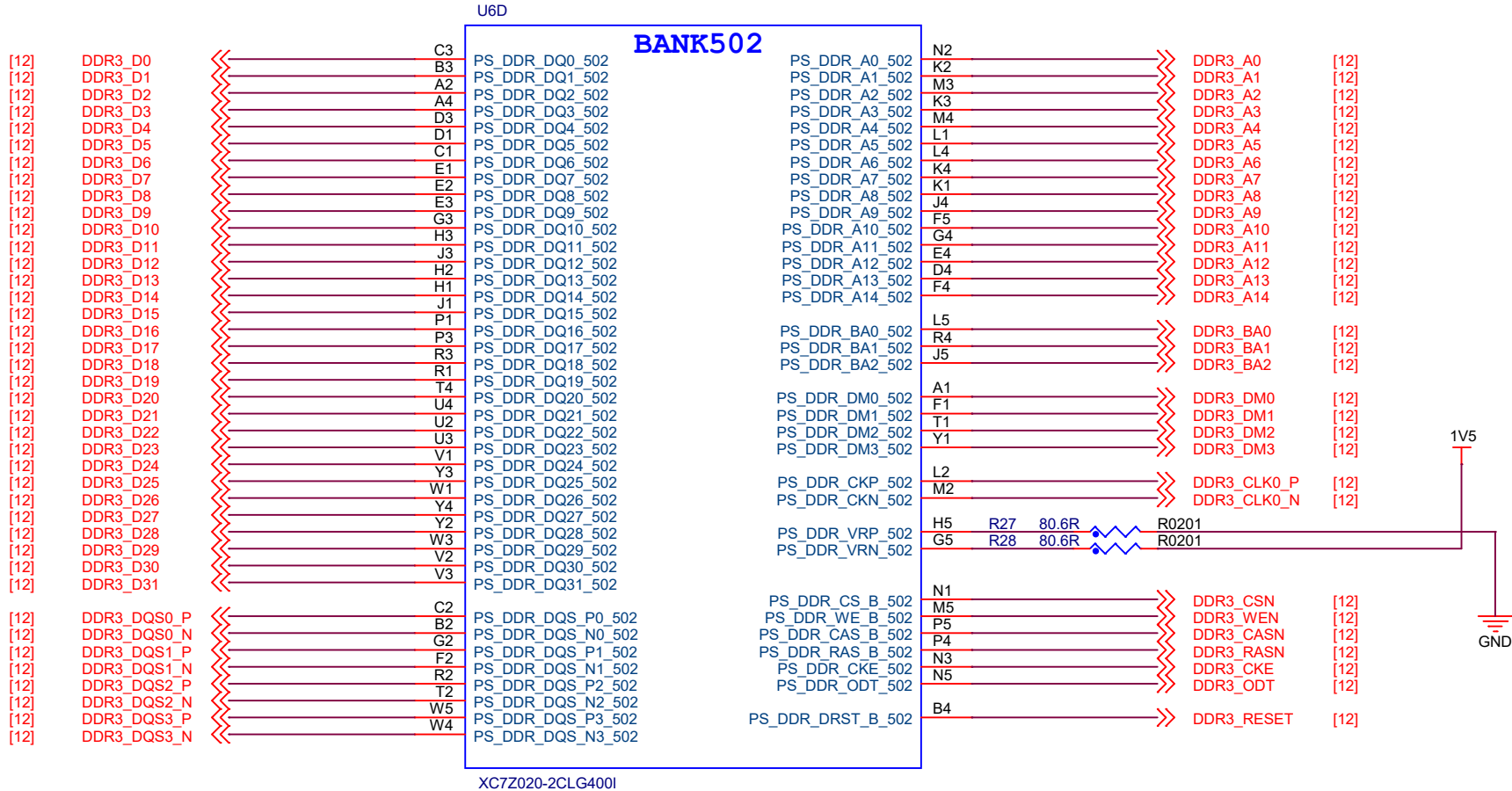
U6C XC7Z020-2CLG400I

BANK501

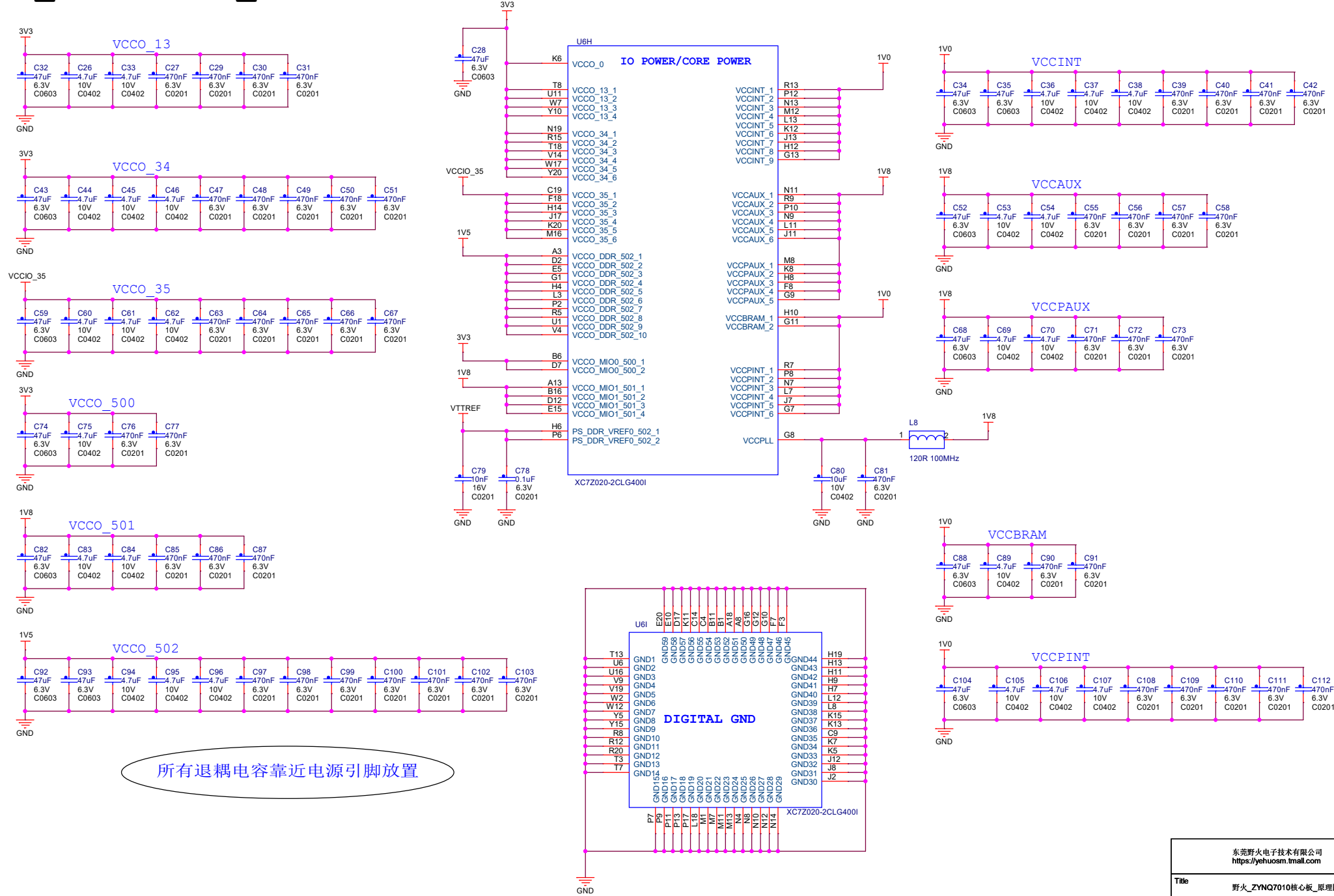


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BANK502



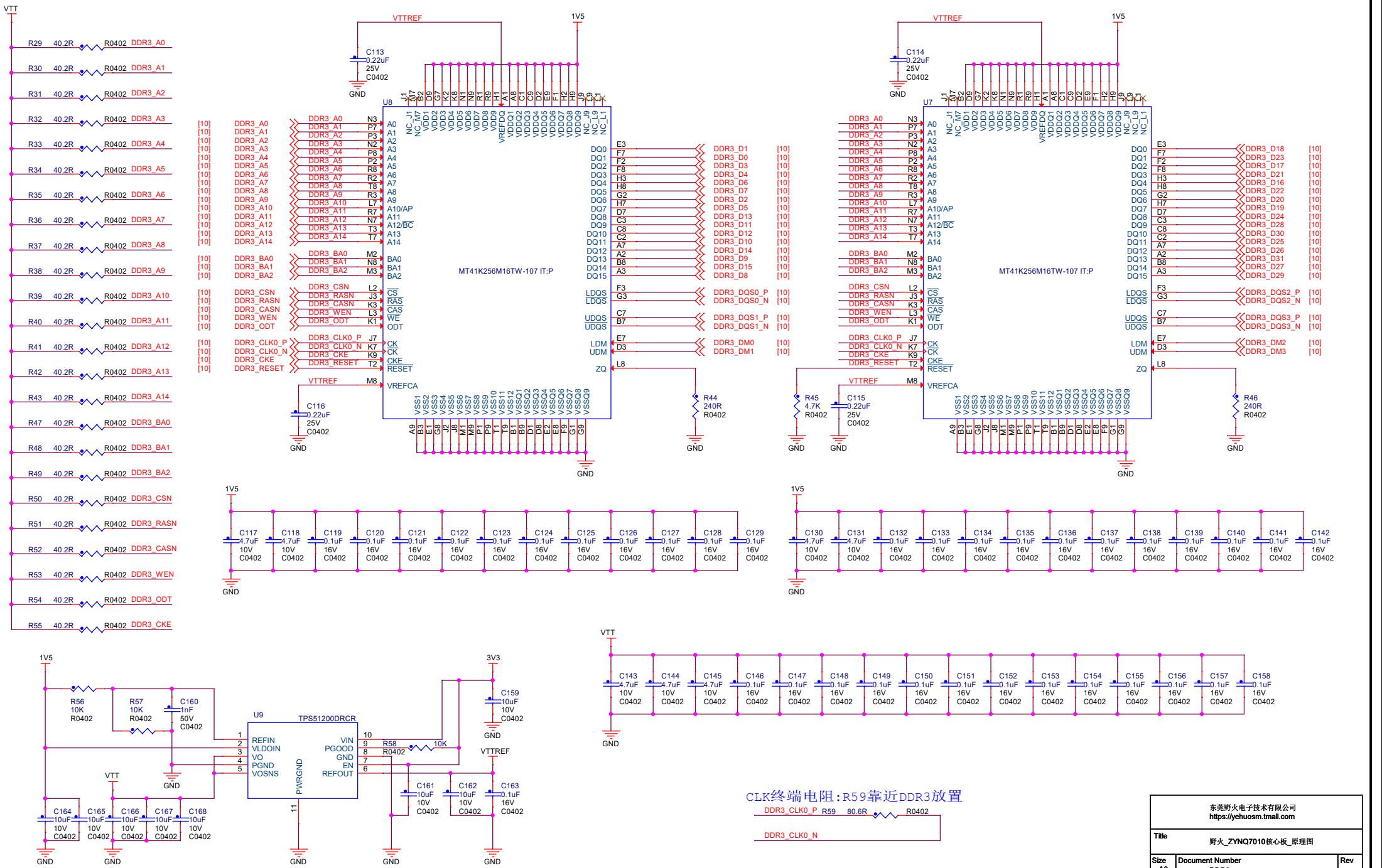
IO_POWER/CORE_POWER



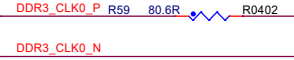
所有退耦电容靠近电源引脚放置

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DDR3



CLK终端电阻:R59靠近DDR3放置



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Table 6-4: Boot Mode MIO Strapping Pins

Pin-signal / Mode	MIO[8]	MIO[7]	MIO[6]	MIO[5]	MIO[4]	MIO[3]	MIO[2]
	VMODE[1]	VMODE[0]	BOOT_MODE[4]	BOOT_MODE[0]	BOOT_MODE[2]	BOOT_MODE[1]	BOOT_MODE[3]
Boot Devices							
JTAG Boot Mode; cascaded is most common ⁽¹⁾				0	0	0	JTAG Chain Routing ⁽²⁾ 0: Cascade mode 1: Independent mode
NOR Boot ⁽³⁾				0	0	1	
NAND				0	1	0	
Quad-SPI ⁽³⁾				1	0	0	
SD Card				1	1	0	
Mode for all 3 PLLs							
PLL Enabled			0	Hardware waits for PLL to lock, then executes BootROM.			
PLL Bypassed			1	Allows for a wide PS_CLK frequency range.			
MIO Bank Voltage ⁽⁴⁾							
	Bank 1	Bank 0	Voltage Bank 0 includes MIO pins 0 thru 15.				
2.5 V, 3.3 V	0	0	Voltage Bank 1 includes MIO pins 16 thru 53.				
1.8 V	1	1					

- Notes:
- 1. JTAG cascaded mode is most common and is the assumed mode in all the references to JTAG mode except where noted.
 - 2. For secure mode, JTAG is not enabled and MIO[2] is ignored.
 - 3. The Quad-SPI and NOR boot modes support execute-in-place (this support is always non-secure)
 - 4. Voltage Banks 0 and 1 must be set to the same value when an interface spans across these voltage banks. Examples include NOR, 16-bit NAND, and a wide TPIU test port. Other interface configuration may also span the two banks.

